

JOSHUA DESGAGNÉS

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PROFILE

Bilingual professional with experience designing databases, building data workflows, and analyzing large datasets. Skilled in relational database design and data integration, with experience developing data systems and supporting reporting, analysis, and operational decision-making. Strong analytical, organizational, and communication skills developed through both technical solutions implementation and client-facing professional experience.

TECHNICAL SKILLS & LANGUAGES

- **Programming:** Competent in SQL (primarily PL/pgSQL), and Python (pandas, psycopg, SQLAlchemy).
- **Web & API Development:** Working knowledge of SQLPage, HTML/CSS, JavaScript (OpenLayers), and FastAPI.
- **Databases:** Competent in PostgreSQL 18/PostGIS 3.6; working knowledge of Azure SQL Database and Microsoft Access; academic exposure to Oracle 19c and Microsoft SQL Server 2017.
- **Data Integration & ETL:** Competent in FME Form, FME Flow, and GDAL/ogr2ogr.
- **Geospatial Platforms & Services:** Working knowledge of pg_featureserv, GeoServer, Martin, GeoNode.
- **Analytics & Visualization:** Proficient in QGIS, ArcGIS Pro, and ArcGIS Dashboards; working knowledge of Apache Superset; academic exposure to Tableau, Power BI, and R.
- **DevOps & Version Control:** Competent in Docker and Git/GitHub.
- **Languages:** Native proficiency in French and English.

WORK EXPERIENCE

COMMUNITY WORKS COLLABORATIVE

Winnipeg, Manitoba

Geospatial Data Engineer

Oct 2025 – Present

- Designed a multi-schema PostgreSQL/PostGIS database supporting 16 subject areas and large-scale spatial datasets, including audit logging, role-based access controls, and data validation workflows.
- Migrated the organization's spatial datasets from Esri file geodatabases and legacy nested folder structures on a shared network drive into a PostgreSQL database using GDAL/ogr2ogr, reorganized by subject domain, replacing proprietary file formats with a centralized database.
- Built a Python ETL pipeline to consolidate Indigenous band membership records from multiple inconsistent Excel sources into a normalized database schema, using a two-hash strategy to distinguish record identity from record content and classify each member per run as new, updated, unchanged, or removed, with field-level change logging, mass-change alerting, a staged soft-delete workflow, and address geocoding via the Nominatim API.
- Developed an internal geoportal platform supporting spatial data search, metadata management, layer discovery, multi-format export, and browser-based map viewing with collaborative project workspaces and non-destructive layer editing using PostGIS, OpenLayers, MapLibre GL, and FastAPI.
- Built a web-based membership registry application (SQLPage) as the primary staff interface to the registry schema, covering individuals, addresses, enrolments, applications, and voting records, with session authentication and role-based access, including an administrative SQL query interface backed by a PL/pgSQL function enforcing read-only execution, table and column validation, and per-query statement timeouts.
- Contributed to documentation covering database standards, data governance, migration procedures, and data management practices.

RENTAL PROTECTION FUND

Vancouver, British Columbia

Geospatial Database Developer Intern

Apr 2025 – May 2025

- Led the design and implementation of a relational database in Microsoft Azure SQL Database using DBEaver, defining schemas, constraints, and views to integrate housing and applicant data.
- Built automated ETL pipelines with FME Form to ingest, validate, and load data from multiple sources.
- Deployed interactive dashboards using Apache Superset for operational reporting and ad hoc querying.
- Translated business requirements into technical specifications, troubleshoot data inconsistencies, and communicated workflows to non-technical stakeholders.

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BOOKE & PARTNERS

Staff Accountant

Winnipeg, Manitoba

Jan 2023 – Aug 2024

- Analyzed complex financial datasets and prepared hundreds of financial statements and tax returns for individuals, corporations, and trusts using Excel, ensuring accuracy, compliance, and strict confidentiality.
- Reconciled client accounts at year-end, performing data validation and preparing adjusting entries to resolve discrepancies.
- Managed multiple concurrent projects under demanding deadlines, communicating complex financial information clearly to clients and team members.

EDUCATION

BRITISH COLUMBIA INSTITUTE OF TECHNOLOGY

Advanced Diploma

Burnaby, British Columbia

May 2025

- Geographic Information Systems (GIS)

<https://joshdesgagnes.github.io/gisportfolio/>

UNIVERSITY OF MANITOBA – ASPER SCHOOL OF BUSINESS

Bachelor of Commerce (Honours)

Winnipeg, Manitoba

Apr 2022

- **Majors:** Accounting, Management Information Systems (MIS)

ACADEMIC PROJECTS

INCIDENT REPORTING WEB APPLICATION

Apr 2025

<https://github.com/joshdesgagnes/experience-builder-app>

- Developed a full-stack web application using React, TypeScript, and ArcGIS Maps SDK for JavaScript, featuring data entry forms with validation and real-time synchronization of submitted records.
- Designed modular, reusable components supporting interactive map visualization, keyword-based filtering, and dynamic reporting of user-submitted data.

INFORMATION SYSTEMS STRATEGY

Apr 2022

- Conducted an analysis of a major Canadian transportation company's information systems, collaborating with the VP of Network Analytics & Development to produce a strategic proposal addressing technology constraints and business alignment.
- Assessed existing systems against operational requirements, identifying gaps and producing recommendations to support a planned strategic expansion.

CHESS DATABASE

Dec 2021

- Designed and built a relational database to store, query, and analyze collections of chess games for player preparation.
- Processed and transformed data from thousands of Portable Game Notation (PGN) files using Python (Pandas), performing data cleaning, normalization, and loading into structured tables for analytical querying.
- Created a GUI in Microsoft Access for data retrieval and analysis (e.g., win percentages, move orders, variation frequency).